

Experiment No. 2

Aim: Write a code to the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file

Theory:

Introduction :

- The find-S algorithm is a basic concept learning algorithm in machine learning.
- The find-S algorithm finds the most specific hypothesis that fits all the positive examples.
- We have to note here that the algorithm considers only those positive training example.
- The find-S algorithm starts with the most specific hypothesis and generalizes this hypothesis each time it fails to classify an observed positive training data.
- Hence, the Find-S algorithm moves from the most specific hypothesis to the most general hypothesis.

Important Representation :

1. ? indicates that any value is acceptable for the attribute.
2. specify a single required value (e.g., Cold) for the attribute.
3. Φ indicates that no value is acceptable.
4. The most **general hypothesis** is represented by: {?, ?, ?, ?, ?, ?}
5. The most **specific hypothesis** is represented by: { Φ , Φ , Φ , Φ , Φ , Φ }

Find-S Algorithm Machine Learning

1. Initialize h to the most specific hypothesis in H
2. For each positive training instance x
 For each attribute constraint a_i in h
 If the constraint a_i is satisfied by x
 then do nothing
 Else
 replace a_i in h by the next more general constraint that is satisfied by x
3. Output the hypothesis h

Example:

Step – 1 of Find-S Algorithm

Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
2	Sunny	Warm	High	Strong	Warm	Same	Yes
3	Rainy	Cold	High	Strong	Warm	Change	No
4	Sunny	Warm	High	Strong	Cool	Change	Yes

1. Initialize h to the most specific hypothesis in H

$$h_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset \rangle$$

Step 2 of Find-S Algorithm First iteration

$$h_0 = (\emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset)$$

$$X_1 = \langle \text{Sunny}, \text{Warm}, \text{Normal}, \text{Strong}, \text{Warm}, \text{Same} \rangle$$

$$h_1 = \langle \text{Sunny}, \text{Warm}, \text{Normal}, \text{Strong}, \text{Warm}, \text{Same} \rangle$$

Step 2 of Find-S Algorithm Second iteration

$$h_1 = \langle \text{Sunny}, \text{Warm}, \text{Normal}, \text{Strong}, \text{Warm}, \text{Same} \rangle$$

$$X_2 = \langle \text{Sunny}, \text{Warm}, \text{High}, \text{Strong}, \text{Warm}, \text{Same} \rangle$$

$$h_2 = \langle \text{Sunny}, \text{Warm}, ?, \text{Strong}, \text{Warm}, \text{Same} \rangle$$

Step 2 of Find-S Algorithm Third iteration

$$h_2 = \langle \text{Sunny}, \text{Warm}, ?, \text{Strong}, \text{Warm}, \text{Same} \rangle$$

$$X_3 = \langle \text{Rainy}, \text{Cold}, \text{High}, \text{Strong}, \text{Warm}, \text{Change} \rangle - \text{No}$$

X_3 is Negative example Hence ignored

$$h_3 = \langle \text{Sunny}, \text{Warm}, ?, \text{Strong}, \text{Warm}, \text{Same} \rangle$$

Step 2 of Find-S Algorithm Fourth iteration

$h_3 = \langle \text{Sunny, Warm, ?, Strong, Warm, Same} \rangle$

$X_4 = \langle \text{Sunny, Warm, High, Strong, Cool, Change} \rangle$

$h_4 = \langle \text{Sunny, Warm, ?, Strong, ?, ?} \rangle$

Step 3

The final maximally specific hypothesis is $\langle \text{Sunny, Warm, ?, Strong, ?, ?} \rangle$

1. After we have completed all the training examples we will have the final hypothesis when can use to classify the new examples.

Code:

To understand the implementation, let us try to implement it to a smaller data set with a bunch of examples to decide if a person wants to go for a walk.

The concept of this particular problem will be on what days does a person likes to go on walk.

Time	Weather	Temperature	Company	Humidity	Wind	Goes
Morning	Sunny	Warm	Yes	Mild	Strong	Yes
Evening	Rainy	Cold	No	Mild	Normal	No
Morning	Sunny	Moderate	Yes	Normal	Normal	Yes
Evening	Sunny	Cold	Yes	High	Strong	Yes

Looking at the data set, we have six attributes and a final attribute that defines the positive or negative example. In this case, yes is a positive example, which means the person will go for a walk.

So now, the general hypothesis is:

$h_0 = \{ \text{'Morning', 'Sunny', 'Warm', 'Yes', 'Mild', 'Strong'} \}$

This is our general hypothesis, and now we will consider each example one by one, but only the positive examples.

$h_1 = \{ \text{'Morning', 'Sunny', '?', 'Yes', '?', '?'} \}$

$h_2 = \{ \text{'?', 'Sunny', '?', 'Yes', '?', '?'} \}$

We replaced all the different values in the general hypothesis to get a resultant hypothesis.

Code:

```
import pandas as pd
import numpy as np

#to read the data in the csv file
data = pd.read_csv("data.csv")
print(data,"n")

#making an array of all the attributes
d = np.array(data)[:,-1]
print("n The attributes are: ",d)

#segragating the target that has positive and negative examples
target = np.array(data)[:,-1]
print("n The target is: ",target)

#training function to implement find-s algorithm
def train(c,t):
    for i, val in enumerate(t):
        if val == "Yes":
            specific_hypothesis = c[i].copy()
            break

    for i, val in enumerate(c):
        if t[i] == "Yes":
            for x in range(len(specific_hypothesis)):
                if val[x] != specific_hypothesis[x]:
                    specific_hypothesis[x] = "?"
            else:
                pass

    return specific_hypothesis

#obtaining the final hypothesis
print("n The final hypothesis is:",train(d,target))
```

Output:

```
Time Weather Temperature Company Humidity Wind Goes
0 Morning Sunny Warm Yes Mild Strong Yes
1 Evening Rainy Cold No Mild Normal No
2 Morning Sunny Moderate Yes Normal Normal Yes
3 Evening Sunny Cold Yes High Strong Yes

The attributes are: [['Morning' 'Sunny' 'Warm' 'Yes' 'Mild' 'Strong']
['Evening' 'Rainy' 'Cold' 'No' 'Mild' 'Normal']
['Morning' 'Sunny' 'Moderate' 'Yes' 'Normal' 'Normal']
['Evening' 'Sunny' 'Cold' 'Yes' 'High' 'Strong']]

The target is: ['Yes' 'No' 'Yes' 'Yes']

The final hypothesis is: ['?' 'Sunny' '?' 'Yes' '?' '?']
```

Conclusion:



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